MSC DATA SCIENCE PROJECT BOOK

# Abstract, Dataset, Features and Daily Work Record

*Student Success Analytics and Early Warning System*

**Prepared by:** [Student Name / Register Number]

**Programme:** MSc Data Science

**Project status:** Easy-language record of completed project work

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**Human review is required. The system supports academic intervention; it does not make decisions about students.**

# 1. Project Abstract

This project is called Student Success Analytics and Early Warning System. The main purpose is to help faculty members identify students who may need academic support at an early stage. The system studies information available during the course, such as assessment submission, marks, previous attempts, study credits, and activity in the online learning system.

The project uses machine learning to estimate the chance of a later Fail or Withdrawn result. It places each record in Low, Medium, or High support risk. These levels are only used to arrange faculty review. They are not final decisions and they do not mean that a student will definitely fail. Faculty members must check the record and talk to the student before choosing any support.

Different models were tested, and a calibrated Random Forest model was selected using validation data. On the separate test data, the model obtained ROC-AUC 0.761 and precision-recall AUC 0.711. The High group found 45.5% of the later adverse outcomes, and 72.5% of High-group records later had an adverse outcome. Medium and High together found 74.5% of adverse outcomes. The project also includes SHAP explanations, fairness checks, support suggestions, and dashboards.

# 2. Where We Got the Data

We used the Open University Learning Analytics Dataset, commonly called OULAD. The original data was published by The Open University and is anonymized. It can be downloaded from the official OULAD source, and copies are also available on Kaggle. If a Kaggle copy is used, the original Open University source must still be mentioned.

The dataset contains seven connected CSV files:

- studentInfo.csv - student, module, previous attempt, credits, and final result information.
- studentRegistration.csv - registration and withdrawal dates.
- assessments.csv - assessment date, type, and weight.
- studentAssessment.csv - student submission date and score.
- studentVle.csv - dated online-learning clicks for each student.
- vle.csv - information about online-learning activities.
- courses.csv - module presentation information.

We joined these files and created one row for one student in one module presentation. We used only the information available up to day 40. The prepared data has 27,042 rows.

## 3. Output We Are Predicting

The model predicts a later adverse academic outcome. Fail and Withdrawn are placed in the adverse group. Pass and Distinction are placed in the successful group. Final results are used only to train and test the model; they are not used as input features.

| Support level | Simple meaning |
|---|---|
| Low | The record has a lower model-estimated need for review. |
| Medium | The record may need a normal academic check-in. |
| High | The record should be reviewed first by faculty. |

## 4. Features We Are Taking

| Feature group | Features used in simple words |
|---|---|
| Previous study | Number of previous attempts and date of registration. |
| Study load | Number of credits studied. |
| Assessment work | Assessments due, submitted, late, and completed by day 40. |
| Assessment marks | Average score, minimum score, weighted score, and submitted assessment weight. |
| Online activity | Total clicks, active days, clicks in the last 7 and 14 days, and recent active days. |
| Activity change | Previous-week clicks, change in clicks, and days since the last activity. |
| Course details | Module code and presentation season. |

Gender, age band, disability, highest education, and deprivation band are not used to make the prediction. They are kept separately only to check whether the model shows different error patterns for different groups.

Important note: OULAD does not have normal classroom attendance, GPA, or backlog columns. We did not create false values for these fields. Online clicks are treated as online engagement, not as classroom attendance.

## 5. Simple Project Method

| Step | Activity |
|---|---|
| 1 | Understand the student-support problem and decide the project scope. |
| 2 | Collect and study the OULAD CSV files. |

| Step | Activity |
|---|---|
| 3 | Clean the data and join the related tables. |
| 4 | Create day-40 assessment and online-activity features. |
| 5 | Check missing values, duplicates, data types, and distributions. |
| 6 | Split the data by time into training, validation, and testing groups. |
| 7 | Train and compare four classification models. |
| 8 | Select model thresholds using validation data only. |
| 9 | Evaluate the final model on untouched test data. |
| 10 | Use SHAP to explain model contributions. |
| 11 | Check fairness indicators for approved groups. |
| 12 | Create faculty, student, and evaluation dashboards. |
| 13 | Prepare Power BI data, measures, theme, and report-page plan. |
| 14 | Test the code and prepare project documents. |

# 6. Day-to-Day Work Record

The following record is written by work session because exact calendar dates were not recorded in the project files. Dates can be added in the left column of the physical project book.

### Day 1 - Topic Selection and Scope

Discussed possible MSc Data Science project topics and selected Student Success Analytics and Early Warning System. Wrote the basic problem statement, main aim, expected users, and project scope. Confirmed that the system will support faculty review and will not make automatic decisions about students.

### Day 2 - Guideline Study

Read the given data analytics and data preprocessing guidelines carefully. Matched the project with problem identification, requirement analysis, data collection, cleaning, preprocessing, modelling, evaluation, dashboard development, and documentation. Prepared a clear workflow so that every required stage would be covered.

### Day 3 - Dataset Selection

Studied different public student-performance datasets available from official sources and Kaggle. Selected the Open University Learning Analytics Dataset because it has dated assessment and online-learning activity records. Recorded that the original source is The Open University, even if a Kaggle mirror is used.

### Day 4 - Project Setup

Created the Python project folders for source code, raw data, processed data, documents, tests, configuration, dashboards, and model outputs. Prepared the virtual environment and requirements file. Added the first project charter, dataset plan, data dictionary, and responsible-use notes.

### Day 5 - Understanding the Raw Data

Collected the seven OULAD CSV files and studied the meaning of their columns. Checked how student information, registration, courses, assessments, submissions, and online-learning activity are connected. Identified the keys required to combine student, module, and presentation records correctly.

### Day 6 - Data Integration

Developed the data-integration process to join the related OULAD tables. Combined student details, registration timing, assessment information, submission records, and VLE activity using the correct identifiers. Created one consistent student-module-presentation record for later feature engineering.

### Day 7 - Data Cleaning and Validation

Checked the raw and combined data for missing values, duplicate keys, invalid ranges, and inconsistent outcomes. Removed records that were not suitable for the selected prediction setup. Added validation checks so that negative counts, impossible rates, and duplicate student-module-presentation rows could be detected.

### Day 8 - Prediction Cutoff and Leakage Control

Selected day 40 of the module presentation as the early prediction point. Excluded students whose withdrawal was already known on or before day 40. Ensured that final results and any information recorded after day 40 were not used as model features, preventing target leakage.

### Day 9 - Assessment Feature Engineering

Created assessment features using only work due by day 40. Calculated assessments due, assessments submitted, late assessments, submitted assessment weight, average score, minimum score, weighted score, and completion rates. Checked that completion rates did not exceed the valid range.

### Day 10 - Online-Activity Features

Processed the large studentVle.csv file in chunks to control memory use. Created total clicks, active days, last 7-day and 14-day clicks, previous-week clicks, recent active days, days since last activity, activity trend, and a no-activity flag. Treated these values as online engagement, not attendance.

### Day 11 - Exploratory Data Analysis

Performed exploratory data analysis on the prepared day-40 dataset. Studied the number of records, modules, presentations, outcomes, missing values, numerical summaries, and engagement distributions. Generated reusable CSV, JSON, and HTML EDA outputs for the report and dashboard.

### Day 12 - Target Definition

Created the binary target called adverse_outcome. Fail and Withdrawn after the cutoff were placed in the adverse group, while Pass and Distinction were placed in the successful group. Documented that this target is an observed historical outcome and not a diagnosis of a student.

### Day 13 - Chronological Data Split

Divided the dataset by academic presentation instead of using a random split. Used 2013B and 2013J for training, 2014B for model validation, and 2014J as the untouched test cohort. This setup gives a more realistic check of performance on later student records.

### Day 14 - Model Training

Built four classification baselines: Dummy Prior, Logistic Regression, Random Forest, and Histogram Gradient Boosting. Applied categorical encoding, missing-value handling, and other preprocessing inside the model pipeline. Trained each model using the training presentations without using the final test data.

### Day 15 - Model Comparison and Thresholds

Compared the models using validation ROC-AUC, average precision, probability error, High-risk recall, High-risk precision, and review rate. Calibrated the model probabilities and selected the Random Forest. Chose Low, Medium, and High thresholds using validation data and assumed faculty review capacity.

### Day 16 - Final Model Evaluation

Evaluated the selected calibrated Random Forest once on the untouched 2014J test cohort. Recorded ROC-AUC 0.761, precision-recall AUC 0.711, High recall 45.5%, High precision 72.5%, and Medium-plus-High recall 74.5%. Clearly stated that these are historical public-data results.

### Day 17 - SHAP Explainability

Generated global and sampled student-level SHAP explanations for the selected model. Identified which available assessment, activity, study-load, and module features contributed most to model estimates. Documented that SHAP explains the model calculation and does not prove the cause of a student's outcome.

### Day 18 - Fairness and Ethics Review

Checked group-level outcome rates, selection rates, recall, precision, and false-positive rates for approved demographic fields. Kept these sensitive fields outside the prediction feature list. Prepared privacy, access-control, correction, retention, human-review, and prohibited-use rules for any future institutional pilot.

### Day 19 - Support and Intervention Rules

Created transparent support indicators from visible academic records, such as incomplete assessments, low available scores, no recent activity, or declining engagement. Mapped each indicator to possible faculty-reviewed support, including tutoring, feedback, access help, study planning, or an adviser check-in. No action is automatic.

### Day 20 - Streamlit Dashboard

Built separate Faculty, Student, and Evaluation views in Streamlit. The Faculty view contains the prioritized review queue, indicators, and support options. The Student view avoids risk labels and focuses on learning records and support, while the Evaluation view presents model results and fairness information.

### Day 21 - Testing and Run Scripts

Added automated tests for schemas, labels, feature engineering, risk tiers, and intervention rules. Corrected the complete PowerShell workflow so that the large prepared dataset can be reused without rebuilding it each time. Verified that all 13 automated tests passed in the latest run.

### Day 22 - Power BI Preparation

Prepared an Excel data model that can be imported into Power BI. Added fact and dimension tables, DAX measures, a colour theme, relationships, and a five-page dashboard plan. Noted that the native .pbix file must be assembled on a computer where Power BI Desktop is installed.

### Day 23 - Final Documentation and Update

Collected the completed work into the project report, model card, ethics protocol, dataset notes, Power BI guide, and progress update. Checked that the project follows the provided analytics guidelines. Listed the remaining work: final university formatting, Power BI assembly, presentation slides, and demonstration practice.

## 7. Main Results

| Item | Result |
|---|---|
| Prepared dataset | 27,042 day-40 student-module records |
| Training data | 11,628 records |
| Validation data | 6,366 records |
| Test data | 9,048 records |
| Selected model | Calibrated Random Forest |
| ROC-AUC | 0.761 |

| Item | Result |
|---|---|
| Precision-recall AUC | 0.711 |
| High recall | 45.5% |
| High precision | 72.5% |
| Medium + High recall | 74.5% |

## 8. Work Completed and Work Remaining

Completed work:

- Dataset collection, cleaning, joining, preprocessing, and EDA.
- Model training, comparison, calibration, testing, SHAP, and fairness audit.
- Support rules, review queue, Streamlit dashboard, Power BI-ready files, and documentation.
- Thirteen automated tests passed in the latest verified run.

Remaining work:

- Open the prepared model in Power BI Desktop and save the native .pbix dashboard.
- Arrange the report in the university's final dissertation format.
- Prepare final presentation slides and demonstration practice.
- Use local student data only after formal approval and new validation; this is not part of the current prototype.

## 9. Conclusion

The project follows the full data analytics workflow: problem selection, requirement study, data collection, understanding, cleaning, preprocessing, visualization, model building, evaluation, interpretation, dashboard development, testing, and documentation. The system helps faculty decide which records to review first, but faculty remain responsible for checking the facts and choosing suitable academic support.

## Reference

Kuzilek, J., Hlosta, M., & Zdrahal, Z. (2017). Open University Learning Analytics Dataset. Scientific Data, 4, 170171. https://doi.org/10.1038/sdata.2017.171